

Mathematical Foundations of Image Synthesis

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1 Introduction

Hello World!!

This is L^AT_EX document.

A rectangle has side lengths of $(x + 1)$ and $(x + 3)$.

The equation of area of the rectangle is

$$A(x) = x^2 + 4x + 3$$

.

2 Mathematical notation

Superscripts :

$$2x^3$$

$$2x^{34}$$

$$2x^{3x+4}$$

$$2x^{3x^4+5}$$

Subscripts:

$$x_1$$

$$x_{12}$$

$$x_{12_2}$$

$$x_{1_{2_3}}$$

$$a_0, a_1, a_2, a_3, \dots, a_{100}$$

Greek Letters:

$$\pi$$

$$\Pi$$

$$\alpha$$

$$A = \pi r^2$$

Trigonometric Functions:

$$y = \sin x$$

$$y = \sin 2x$$

$$y = \sin^2 2x$$

$$y = \csc \theta$$

$$y = \csc \Theta$$

$$y = \sin^{-1} x$$

$$y = \arcsin x$$

Log Functions:

$$y = \log x$$

$$y = \log_4 x$$

$$y = \ln(x^3 + 5)$$

Roots:

$$\sqrt{x^3 + 5}$$

$$\sqrt[4]{3x^7 + 2x + 7}$$

$$\sqrt{\sqrt[3]{1 + \sqrt{x^2 + 2x + 5}}}$$

Fractions:

$$\frac{6}{7}$$

About $\frac{2}{3}$ of the glass is full.

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$$\frac{2}{3}$$

$$2$$

$$\frac{\sqrt{x+1}}{\sqrt{x+2}}$$

$$\frac{1}{1+\frac{1}{\sqrt{x+2}}}$$

3 Bracket,Tables and Arrays

The distributive property states that $a(b+c)=ab+ac$ for all $a,b,c\in\mathbb{R}$.

The equivalence class of a is $[a]$.

The set A is defined to be $\{1,2,3\}$.

The movie ticket costs \$11.50.

$$2\left(\frac{1}{x^2-1}\right)$$

$$2\left\langle \frac{1}{x^2-1}\right\rangle$$

$$2\left|\frac{1}{x^2-1}\right|$$

$$\left.\frac{dy}{dx}\right|_{x=1}$$

$$\left(\frac{1}{1+\left(\frac{1}{1+x}\right)}\right)$$

Tables:

x	1	2	3	4	5
$f(x)$	10	11	12	13	14

x	1	2	3	4	5
$f(x)$	$\frac{1}{2}$	11	12	13	14

Table 1: Practice Tables